DATA SCIENCE LAB 1: ASSIGNMENT 3

CLUSTERING AND FITTING REPORT

Clustering and Fitting Analysis of Economic and Environmental Data

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Data Source: https://data.worldbank.org/
GDP Per Capita: https://data.worldbank.org/indicator/NY.GDP.PCAP.CD
CO_2 Emissions: https://data.worldbank.org/indicator/EN.GHG.CO2.PC.CE.AR5
Git-Repository: https://github.com/Ruzen-Maharjan/DSlab-3.git

Abstract

This report explores the relationship between economic development and environmental impact using World Bank data. Countries were grouped using clustering based on GDP per capita and CO_2 emissions per capita. In addition, curve fitting was used to analyze trends in emissions over time for a selected country. The results show that countries with higher GDP generally have higher emissions, and emissions tend to increase over time.

Introduction

The aim of this report is to analyse differences between countries in terms of economic development and environmental impact. Two methods were used: clustering and curve fitting. Clustering was used to group countries with similar characteristics, while curve fitting was used to study how emissions change over time.

Clustering Analysis

GDP per capita and CO_2 emissions per capita were selected as the main indicators. These were chosen because they represent economic activity and environmental impact.

Before applying clustering, the data was normalized to ensure that both variables contributed equally. The clustering was then performed using the KMeans algorithm. After clustering, the results were displayed using the original values, and the cluster centers were converted back to the original scale. The predicted CO_2 emissions for 2030 are approximately 2.52, with a lower bound of around 2.49 and an upper bound of around 2.56.

Regional aggregates such as "World" and "Africa Eastern and Southern" were removed to ensure that only individual countries were analyzed.

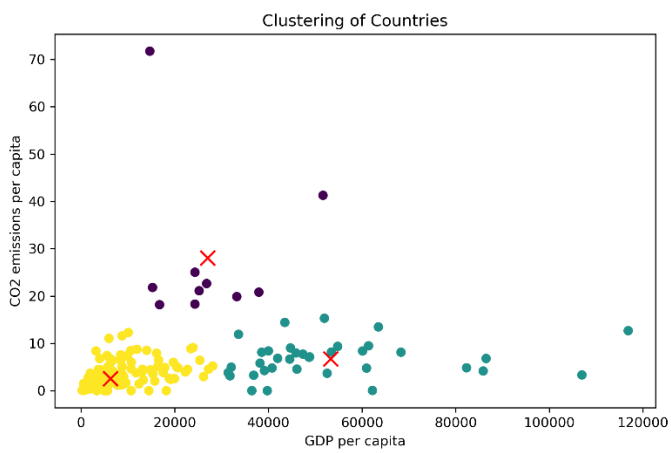

Fig 1: Clustering of countries based on GDP per capita and CO_2 emissions per capita

The results show three main clusters. One cluster contains countries with low GDP and low emissions. Another cluster contains countries with medium GDP and moderate emissions. The final cluster includes countries with high GDP and generally higher emissions.

This suggests that countries with higher economic development tend to produce more CO_2 emissions, although there are some exceptions.

Curve Fitting

To analyze trends over time, India was selected as a case study. CO_2 emissions per capita were analyzed over multiple years.

A simple quadratic model was used to fit the data. This type of model was chosen because it is simple and suitable for capturing gradual changes over time.

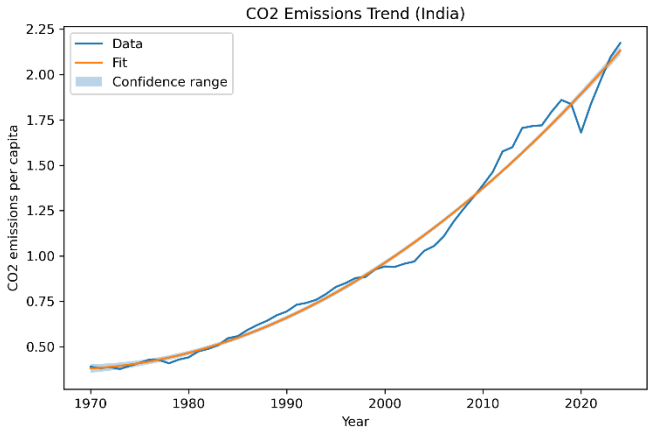

Fig 2: Quadratic fit of CO_2 emissions per capita for India with confidence range

The fitted curve follows the general trend of the data. A confidence range was calculated using the provided error function, which shows the uncertainty in the model.

The model was then used to predict CO_2 emissions in 2030. The results suggest that emissions will continue to increase. However, the confidence range becomes wider in future years, indicating that predictions are less certain further ahead.

Discussion

The clustering results show a clear relationship between economic development and emissions. Countries with higher GDP tend to have higher emissions, likely due to increased industrial activity and energy use.

The curve fitting results support this observation. The trend for India shows that emissions are increasing over time. However, the uncertainty in the prediction suggests that future emissions could vary depending on factors such as policy changes or technological improvements. It can also be observed that not all countries follow the same pattern, suggesting that factors such as renewable energy use and environmental policies may influence emissions.

Conclusion

In conclusion, this analysis shows that there is a relationship between economic growth and environmental impact. Clustering helped identify patterns between countries, while curve fitting provided insight into trends over time and possible future values.